

Karan Gupta

Embedded Systems & IoT Firmware Engineer • 5 Years Experience • MERIQ Index: 74% (Developing)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: Python, MQTT, I2C/SPI/UART, Embedded Linux, C/C++, FreeRTOS, ARM Cortex

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Intel* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core Python service, reducing latency by 37%.
- Led a squad of 4 engineers delivering scalable microservices processing 15M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 44%.

Associate Embedded Systems & IoT Firmware Engineer — *Bosch* (2021 - 2023)

- Designed and maintained MQTT data endpoints with 95% SLA uptime.
- Optimized database indexing and caching layers using I2C/SPI/UART saving \$21k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [Python, MQTT, I2C/SPI/UART, Embedded Linux]

Engineered end-to-end production solution leveraging Python and MQTT with full test automation.

Automotive CAN-Bus Telemetry Gateway with Embedded Linux and Secure Boot [I2C/SPI/UART, Embedded Linux, C/C++, FreeRTOS]

Architected high-resilience system integrating I2C/SPI/UART with 99.9% uptime.

EDUCATION & CREDENTIALS

B.Tech in Computer Science and Engineering

ETH Zurich — 2017 - 2021 (Score: 9.19 / 10.0)